

$$TF = \frac{\text{Number of times a word "X" appears in a Document}}{\text{Number of words present in a Document}}$$
$$IDF = \log \left(\frac{\text{Number of Documents present in a Corpus}}{\text{Number of Documents where word "X" has appeared}} \right)$$
$$TF\text{ IDF} = TF \cdot IDF$$

Document	Text
D1	students watch youtube
D2	youtube watch youtube
D3	students write comments
D4	youtube write comments

students	watch	youtube	write	comments
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TF-IDF intuition

Think of TF-IDF as a way to answer this question:

“How important is a word in a document, compared to all other documents?”

1. TF (Term Frequency)

“How often does the word appear in THIS document?”

- If a word appears many times → it might be important *in this document*
 - Example:
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 - “football football football”
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 - → TF is high for “football”
- So TF = **local importance**

2. IDF (Inverse Document Frequency)

“How rare is this word across ALL documents?”

- Common words like “the”, “is”, “and” → not useful
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 - Rare words → more meaningful
- So IDF = **global importance**

Meaning:

A word is important if it appears a lot in one document AND is rare in others.